Retail Demand Estimation using Public Spatial Data

An Equilibrium Configuration Approach

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Motivation

Retail Demand Estimation & Merger Simulation

Typical Application:

- Retail chain mergers/store acquisitions
  - Price effects
  - Distribution efficiencies
  - Store divestiture/branding remedies

Typical Model:

- Store-level retail demand system
  - Household discrete choice of store

Typical Estimator:

- Requires proprietary data
  - Store-level revenues
  - Store-level quantity shares
  - Store-level prices
  - Consumer-level microdata

- Competition agency implementation:
  - Costly/unavailable data ✗
  - Confidentiality-restricted findings ✗

This Paper's Approach:

- Requires only public spatial data
  - Store locations
  - Distribution center locations
  - Household locations & demography
  - Retail precinct locations & sizes

- Competition agency implementation:
  - Free, available data ✓
  - Publishable findings ✓

Related Literature:

- Store-level retail demand estimation ▶
- Retail entry in continuous geographic space
- Profit inequality estimation of entry games
- Gowrisankaran & Krainer (2011 RJE)

Existing Papers:

- Store-level quantity/revenue/price data ✗
  - Thomadsen (2005 RJE; 2007 MS)
  - Davis (2006 RJE)
  - Manuszak (2010 IJIO)
  - Ho & Ishii (2011 IJIO)
  - Houde (2012 AER)
  - Seim & Waldfogel (2013 AER)
  - Aguirregabiria & Vicentini (2016 JIE)
  - Ellickson, Grieco & Khvastunov (2020 RJE)
- Consumer-level store choice data ✗
  - Chernew, Gowrisankaran & Fendrick (2002 JHE)
  - Gaynor, Kleiner & Vogt (2003 RJE; 2013 JIE)
  - Smith (2004 RES; 2006 RJE)
  - Klopack (2024 RJE)

This Paper:

- Public store location data ✓

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- Store-level retail demand estimation
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Existing Papers:

- Reduced form variable profit functions ✗
  - Seim (2006 RJE)
  - Zhu & Singh (2009 QME)
  - Datta & Sudhir (2013 QME)
  - Orhun (2013 QME)
  - Caoui, Hollenbeck & Osborne (2025)
- Conditional on pre-estimated demand system ✗
  - Chernew, Gowrisankaran & Fendrick (2002 JHE)
  - Aguirregabiria & Vicentini (2016 JIE)

This Paper:

- Microfounded demand system ✓
- Directly estimated ✓

Related Literature:

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- Reduced form variable profit functions ✗
  - Ciliberto & Tamer (2009 E)
  - Ellickson, Houghton & Timmins (2013 RJE)
- Conditional on pre-estimated demand system ✗
  - Crawford & Yurukoglu (2012 AER)
  - Eizenberg (2014 RES)
  - Pakes, Porter, Ho & Ishii (2015 E)
  - Berry, Eizenberg & Waldfogel (2016 RJE; 2016 JIE)
  - Aguirregabiria, Clark & Wang (2016 RJE)
  - Kuehn (2018 RJE; 2020 JIE)
  - Wollman (2018 AER)
  - Fan & Yang (2020 AEJM; 2025 JPE)

This Paper:

- Direct estimation of demand system ✓

Related Literature:

- Store-level retail demand estimation
- Retail entry in continuous geographic space
- Profit inequality estimation of entry games
- Gowrisankaran & Krainer (2011 RJE) ▶
  - Microfounded, store-level demand system ✓
  - Continuous geographic space ✓
  - Endogenous store presences ✓
  - Public store location data ✓

Gowrisankaran & Krainer (2011 RJE):

- Regulated, market-level prices ✗

This Paper:

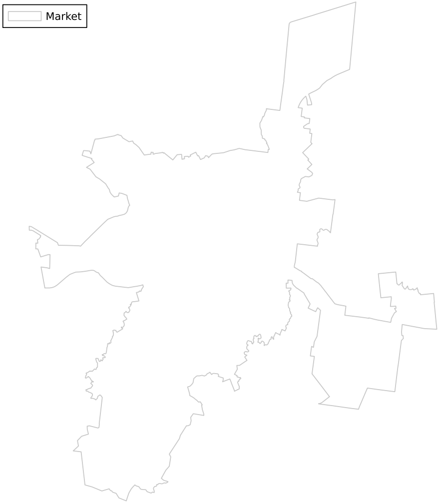
- Profit-maximizing, store-level prices ✓
  - Localized competitive price effects ✓

Model

Data Requirements & Notation

Data Requirements:

- Markets: $m \in \mathcal{M}$
  - Location set: $\mathcal{L}_m$
- Households types: $h \in \mathcal{H}$
  - Income: i_h
  - Location: l_h
  - Population: n_h
- Retail shopping precincts: $r \in \mathcal{R}$
  - Location set: $\mathcal{L}_r$
  - Agglomeration: a_r
- Stores: $s \in \mathcal{S}$
  - Location: l_s
  - Brand: b_s
  - Owner: o_s
- Wholesale distribution centres: $w \in \mathcal{W}$
  - Location: l_w
  - Brand(s) supplied: $\mathcal{B}_w$



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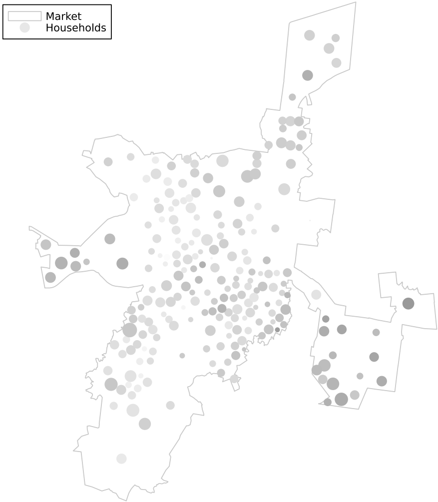

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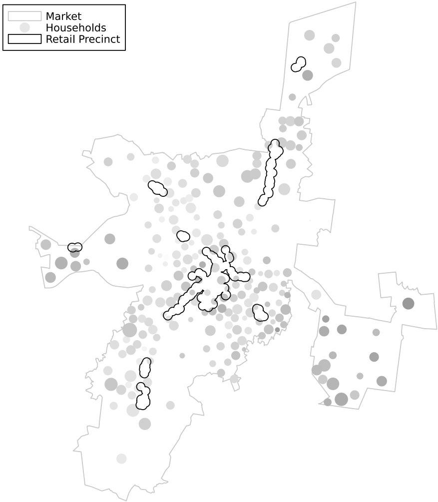

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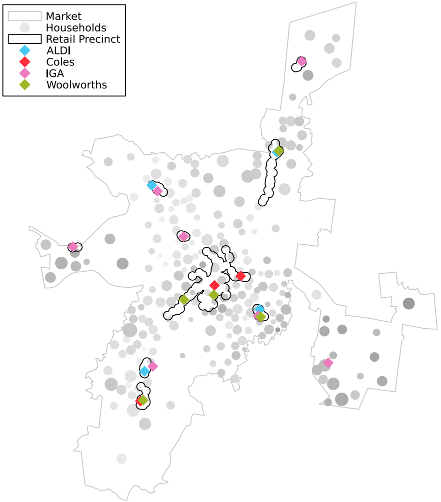

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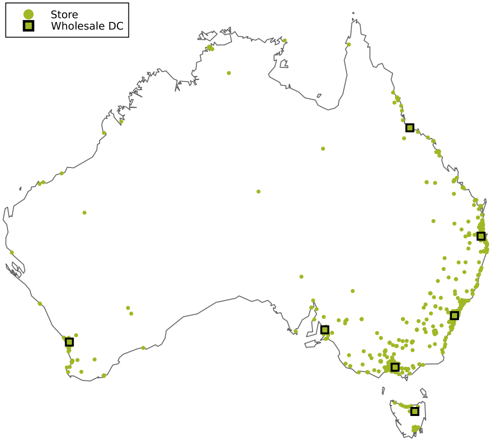


Model

Demand & Price Equilibrium

Household Store Choice

- Brand differentiation
  - Horizontal: ρ
  - Vertical: β_b
- Location differentiation
  - Distance: δ
  - Agglomeration: α
- Price disutility
  - $f(\text{income})$: η_0, η_i

$$u_{hs} = \underbrace{\alpha a_r + \beta_b + \delta d_{hs} + \eta_h p_s}_{\overline{u}_{hs}} + \underbrace{\epsilon_{hb} + (1-\rho)\epsilon_{hs}}_{\nu_{hs}}$$

$$\eta_h = \eta_0 + \eta_i i_h \text{ and } \mathrm{Corr}[\nu_{hs}, \nu_{hs'}] = \rho \text{ for } s, s' \in \mathcal{S}_b$$

$$\underbrace{\psi_{hs}}_{\Pr[s_h = s]} = \underbrace{\frac{\exp(\frac{\overline{u}_{hs}}{1-\rho})}{\Sigma_{bh}}}_{\psi_{hs|b}} \cdot \underbrace{\frac{{\Sigma_{bh}}^{1-\rho}}{1 + \sum_b {\Sigma_{bh}}^{1-\rho}}}_{\psi_{hb}} \text{ with } \Sigma_{bh} = \sum_{s \in \mathcal{S}_b} \exp(\frac{\overline{u}_{hs}}{1-\rho})$$

Owner Profits

- Marginal costs
  - $f(\text{brand, distance})$: μ_b, ω
- Fixed costs
  - $f(\text{brand})$: F_b

$$\pi_o = \sum\nolimits_{s \in \mathcal{S}_o} \pi_s \text{ with } \pi_s = (p_s - c_s)q_s - f_s \text{ and } \qquad q_s = \sum\nolimits_h n_h \psi_{hs}$$

$$c_s = \mu_b + \omega d_{sw}$$

$$\Pr[f_s < x] = F_b(x)$$

Price Equilibrium

- Nash-Bertrand

$$\boldsymbol{p}_o^* = \arg\max_{\boldsymbol{p}_o} \pi_o(\boldsymbol{p}_o, \boldsymbol{p}_{-o}) \quad \forall o \in \mathcal{O} \text{ with } \qquad \boldsymbol{p}_o = [p_s]_{s \in \mathcal{S}_o}$$

Spatial Equilibrium

Sutton (1997 RJE; 1998; 2000; 2007):

Equilibrium configuration

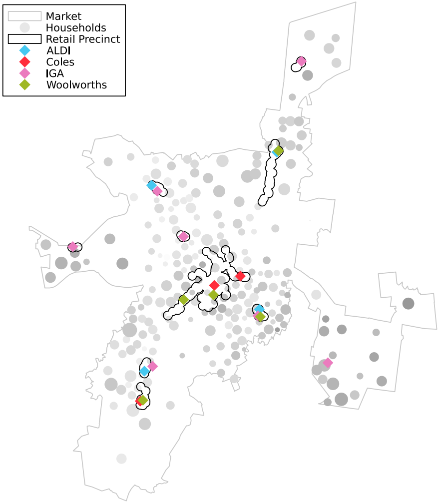
- Mutual best response in store networks
  - Testable marginal profit inequalities:

    $$\Delta\pi_s := \pi_o[s \in \mathcal{S}|\mathcal{S}_{-s}] - \pi_o[s \notin \mathcal{S}|\mathcal{S}_{-s}]$$

    - Observed stores: marginally profitable

      $\Delta\pi_s \geq 0 \quad \forall s \in \mathcal{S}$
    - Unobserved stores: marginally unprofitable

      $\Delta\pi_s \leq 0 \quad \forall s \notin \mathcal{S}$
- Agnostic to order of entry, equilibrium selection
  - Robust to first mover advantages, multiple equilibria
- Long run interpretation
  - Iterated store network updating



Identification

Fixed & Variable Profit Decomposition

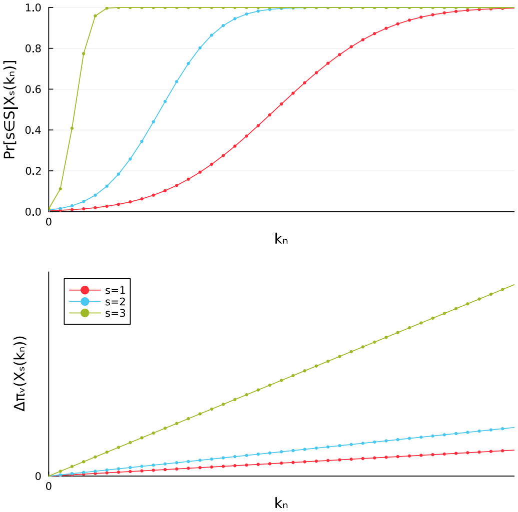
- Conditional store existence probabilities:

$$\Pr[s \in \mathcal{S} | \boldsymbol{\mathcal{X}}_s] = \Pr[\Delta\pi_v(\boldsymbol{\mathcal{X}}_s) - f_s \geq 0 | \boldsymbol{\mathcal{X}}_s]$$
$$= F_b(\Delta\pi_v(\boldsymbol{\mathcal{X}}_s))$$

  - Store-level entry conditioning set: $\boldsymbol{\mathcal{X}}_s = (b_s, l_s, o_s, \mathcal{S}_{-sm}, \mathcal{H}_m, \mathcal{R}_m, \mathcal{W})$
  - Conditional independence assumption: $F_b(\cdot | \boldsymbol{\mathcal{X}}_s) = F_b(\cdot)$

Matzkin (1992 E; 1994), Berry & Tamer (2006):

- Fixed $F_b(\cdot)$ & variable $\Delta\pi_v(\cdot)$ profit decomposition:
  - Shape restriction:
    - Household count multiplier: $\mathcal{H}_m(k_n) = \{h \in \mathcal{H}_m : (i_h, l_h, k_n n_h)\}$
    - Variable profit proportionality: $\Delta\pi_v(\boldsymbol{\mathcal{X}}_s(k_n)) = k_n \Delta\pi_v(\boldsymbol{\mathcal{X}}_s) \propto k_n$
  - Special regressor:
    - $\Delta\pi_v(\boldsymbol{\mathcal{X}}_s(d_{sw}))$ monotone in dense support d_{sw}
  - Scale normalization



Identification

Scale Normalization - Latent Price Units

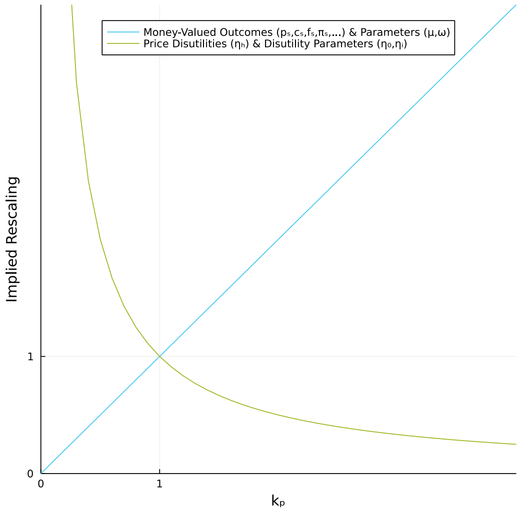
- Conditional store existence probabilities identify:
  - Fixed cost distributions: $F_b(\cdot)$
  - Variable profit parameters: $\boldsymbol{\theta}_v = (\alpha, \delta, \rho, \boldsymbol{\beta}', \eta_0, \eta_i, \omega, \boldsymbol{\mu}')$

  up to a price unit (k_p) scale normalization:

$$F_b\left(\Delta\pi_v\left(\boldsymbol{\mathcal{X}}_s; (\alpha, \delta, \rho, \boldsymbol{\beta}', \eta_0, \eta_i, \omega, \boldsymbol{\mu}')\right)\right) =$$
$$F_b\left(k_p^{-1}\Delta\pi_v\left(\boldsymbol{\mathcal{X}}_s; (\alpha, \delta, \rho, \boldsymbol{\beta}', k_p^{-1}\eta_0, k_p^{-1}\eta_i, k_p\omega, k_p\boldsymbol{\mu}')\right)\right)$$

- Price unit dependent:
  - Store prices: $k_p p_s$
  - Store marginal costs: $k_p c_s$
  - Store fixed costs: $k_p f_s$
  - Store profits: $k_p \pi_s$
  - Household welfare: $k_p W_h$

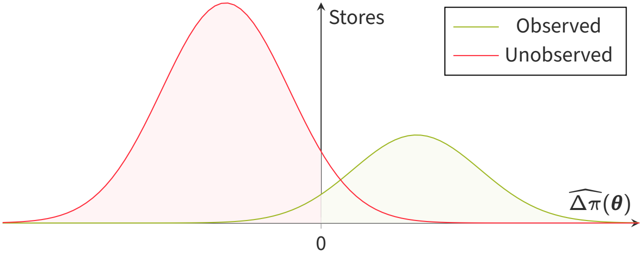
- Scale normalization: $\min_b\{\mu_b\} = 1$
  - Price unit: minimum marginal cost intercept



Estimator:
Maximize weighted log-likelihood of conditional store existence & non-existence probabilities:

$$\ell(\boldsymbol{\theta}) := |\mathcal{S}|^{-1} \sum_{s \in \mathcal{S}} \log \Big[\underbrace{F_b\big(\Delta\pi_v(\boldsymbol{\mathcal{X}}_s; \boldsymbol{\theta}_v); \boldsymbol{\theta}_f\big)}_{\Pr[s \in \mathcal{S}]} \Big] + |\mathcal{B} \times \mathcal{R}|^{-1} \sum_{s \in \mathcal{B} \times \mathcal{R}} \log \Big[\underbrace{1 - F_b\big(\Delta\pi_v(\boldsymbol{\mathcal{X}}_s; \boldsymbol{\theta}_v); \boldsymbol{\theta}_f\big)}_{\Pr[s \notin \mathcal{S}]} \Big]$$

- Outcome-based sampling:
  - $|\mathcal{S}|$ observed stores
  - $|\mathcal{B}| \times |\mathcal{R}|$ unobserved stores
    - Extra store of each brand at each retail precinct
- Parametric fixed cost assumption:
  - $F_b : f_b \sim N(\phi_b, \sigma)$
    - Brand-specific means: ϕ_b
    - Common shocks: $N(0, \sigma)$
  - $\boldsymbol{\theta}_f = (\sigma, \boldsymbol{\phi}')$



Equilibrium Price Computation

Adapted Morrow & Skerlos (2011 OR):

- Fixed point iteration:

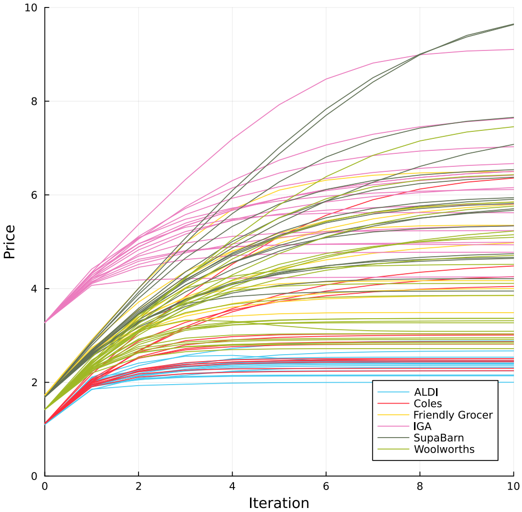
$$\boldsymbol{p} \leftarrow \boldsymbol{c} + \boldsymbol{\Lambda}(\boldsymbol{p})^{-1}\Big((\boldsymbol{O} \odot \boldsymbol{\Gamma}(\boldsymbol{p}))\,(\boldsymbol{p} - \boldsymbol{c}) - \boldsymbol{q}(\boldsymbol{p}) \Big)$$

- Analytic demand derivative matrix decomposition:

$$\boldsymbol{\nabla} = \left[\frac{\partial \pi_i}{\partial p_j}\right]_{S\times S} = \boldsymbol{\Lambda} - \boldsymbol{\Gamma} \text{ with}$$

$$\Lambda_{ii} = \frac{1}{1-\rho} \sum_h n_h \eta_h \psi_{hi}$$

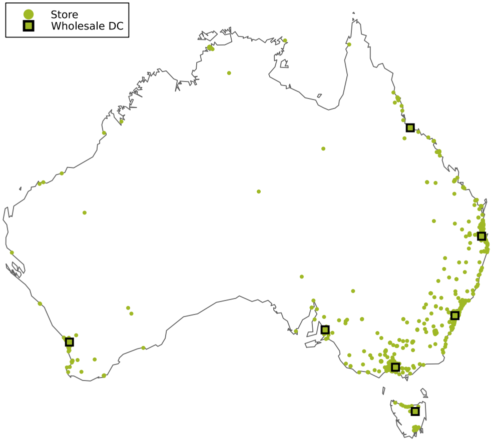
$$\Gamma_{ij} = \sum_h n_h \eta_h \psi_{hi} \psi_{hj} (1 + 1[b_i = b_j] \frac{\rho}{1-\rho} \frac{1}{\psi_{hb}})$$



Data

Full Sample - Australian Supermarkets

- $|\mathcal{M}| \approx$ 800 markets
    - ASGS 2021 Urban Centres & Localities
        - Defined based on contiguous urban/residential density
- $|\mathcal{H}| \approx$ 700,000 household types
    - 2021 Australian census
        - i_h: ln(Weekly household income)
        - l_h: SA1 centroid
        - n_h: Population
- $|\mathcal{R}| \approx$ 2,000 shopping precincts
    - OpenStreetMap
        - $\mathcal{L}_r$: Buffer around density-based shop clusters (DBScan)
        - a_r: ln(Number of shops in cluster)
- $|\mathcal{S}| \approx$ 4,500 supermarkets
    - Brand websites
- $|\mathcal{B}| =$ 11 brands



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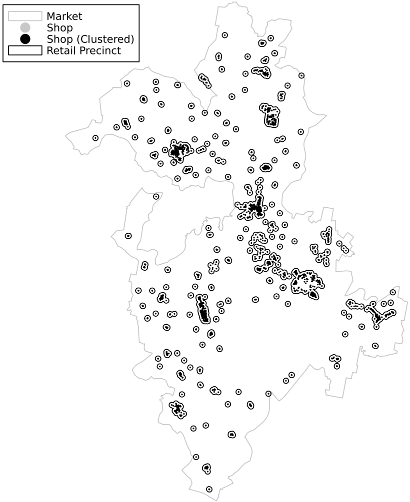

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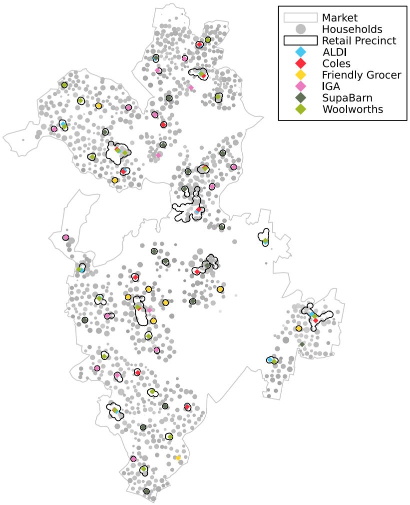
Brand	Stores	Markets	DCs
ALDI	609	139	5
Coles	865	190	6
CostCo	15	8	1
Drakes	65	25	2
FoodLand	90	46	1
IGA	1264	519	6
Woolworths	1134	256	7
FoodWorks	319	181	5
Friendly Grocer	205	94	3
SPAR	110	66	1
SupaBarn	21	3	1

Data

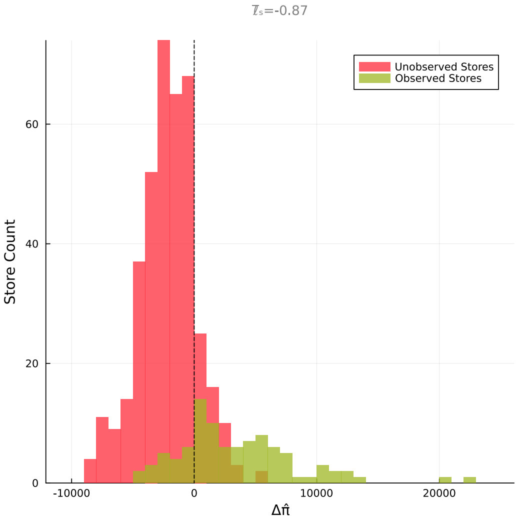
Subsample - Canberra-Queanbeyan

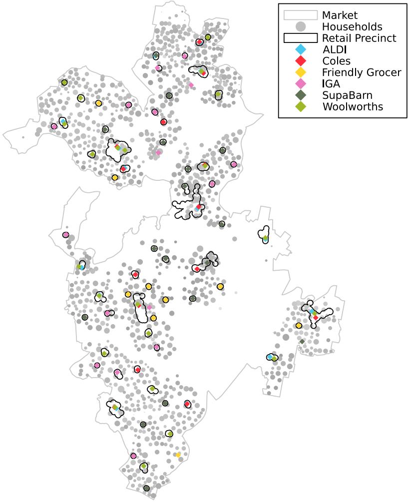
- $|\mathcal{M}| = 1$ market
- $|\mathcal{H}| = 4,494$ household types
  - $\sum_h n_h = 340,770$ total population
  - $|\mathcal{L}_h| = 1,194$ household locations
  - $|\mathcal{I}_h| = 6$ household income bins
- $|\mathcal{S}| = 94$ observed stores
- $|\mathcal{R}| = 65$ shopping precincts
- $|\mathcal{B}| = 6$ brands
  - ALDI
  - Coles
  - Friendly Grocer
  - IGA
  - SupaBarn
  - Woolworths
- $|\mathcal{S}| + |\mathcal{B}||\mathcal{R}| = 484$ profit inequalities



Estimates

Subsample - Canberra-Queanbeyan - Profit Differences



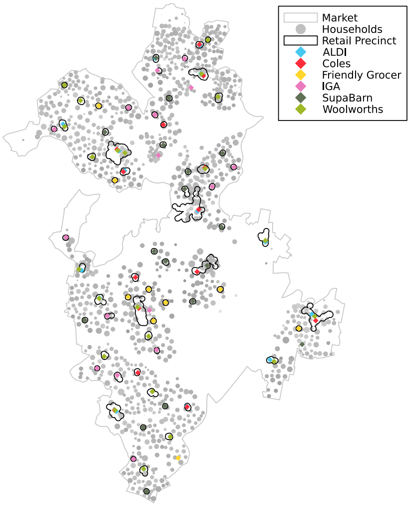


Estimates

Subsample - Canberra-Queanbeyan - Brand Summaries

b	$\bar{p}_s$	$\bar{c}_s$	$\bar{q}_s$	S	$Q^{-1}\Sigma q_s$
ALDI	2.40	1.00	3009	15	0.14
Coles	3.26	1.02	4416	14	0.19
Friendly Grocer	4.92	1.57	2341	9	0.06
IGA	6.11	2.98	1999	17	0.10
SupaBarn	6.21	1.53	3809	17	0.20
Woolworths	4.21	1.29	4522	22	0.30

- Lowest marginal costs: ALDI, Coles
- Highest prices: SupaBarn, IGA
- Highest store volumes: Coles, Woolworths

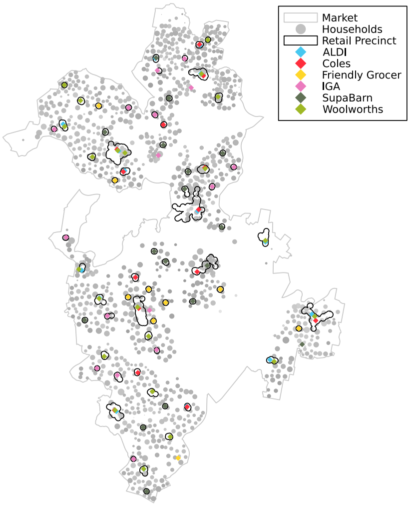


Estimates

Subsample - Canberra-Queanbeyan - Parameter Estimates

α	δ	η_0	η_i	ρ	ω
0.46	-4.12	-4.89	0.49	0.51	0.00

b	β	μ	ϕ
ALDI	10.47	1.00	3114
Coles	11.26	1.02	5450
Friendly Grocer	11.79	1.57	5887
IGA	12.51	2.98	3086
SupaBarn	12.93	1.53	8380
Woolworths	11.99	1.29	4759



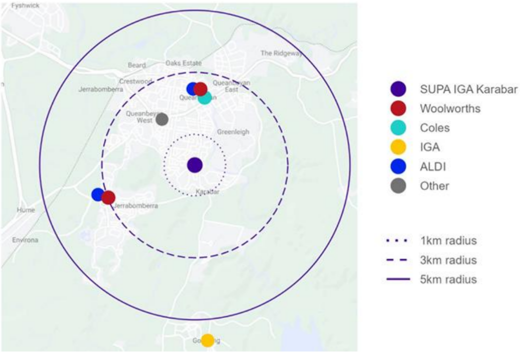
Counterfactuals

Woolworths-IGA Karabar - ACCC (2023)

ACCC (2023)

- Opposed Woolworths acquisition of IGA Karabar
- Geographic market definition:
  - 3-5km radius
  - Exclude IGA Bogong
- Product market definition:
  - 'Full-line' supermarkets
  - Exclude Friendly Grocer ('convenience')
- Counterfactual:
  - Status quo
    - No IGA exit
    - No alternative purchaser
- Substantial lessening of competiton:
  - Concentration: Woolworths 3/6 'full line' supermarkets
  - Differentiation:
    - Woolworths rebrand
    - Loss of differentiated IGA offering

Figure 1: Map of supermarkets in Karabar and surrounding suburbs



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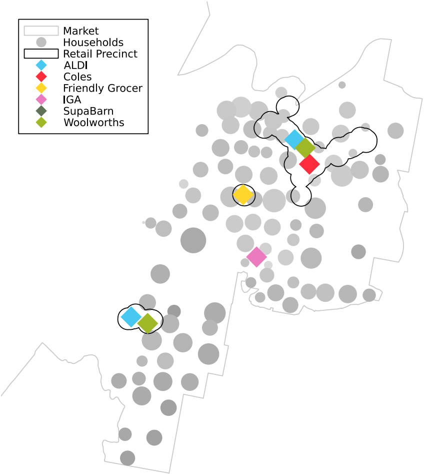
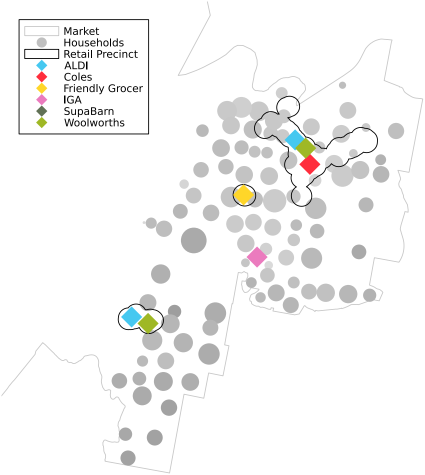


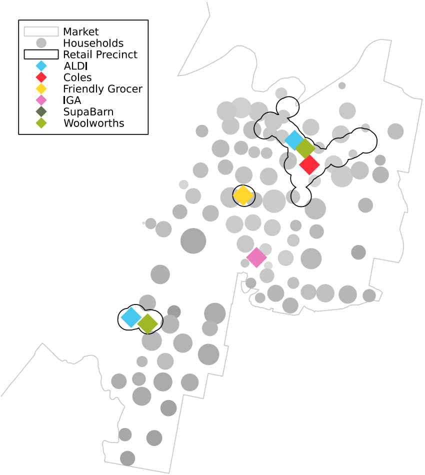
Counterfactuals

Woolworths-IGA Karabar - Merger Counterfactuals

Pre: IGA Karabar



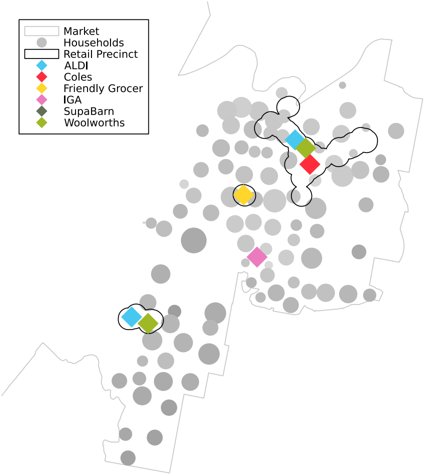
Post: [IGA Woolworths SupaBarn] Karabar



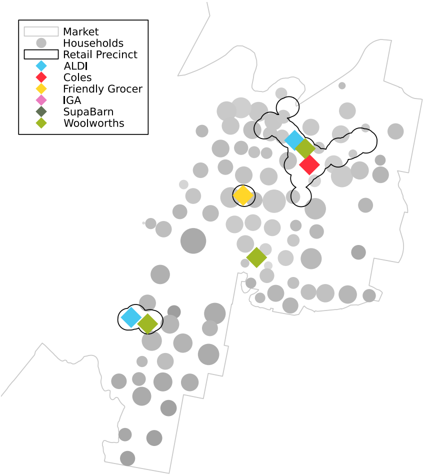
Counterfactuals

Woolworths-IGA Karabar - Merger Counterfactuals

Pre: IGA Karabar



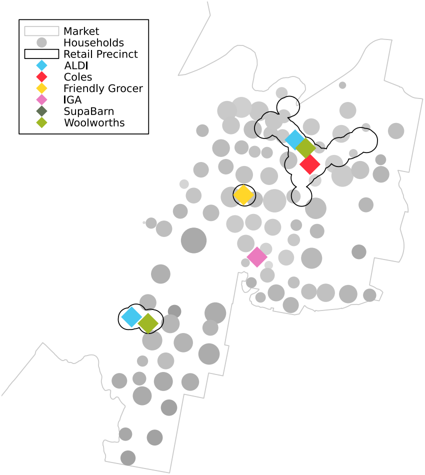
Post: [IGA Woolworths SupaBarn] Karabar



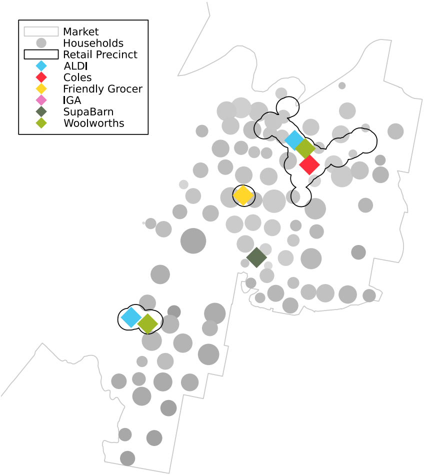
Counterfactuals

Woolworths-IGA Karabar - Merger Counterfactuals

Pre: IGA Karabar

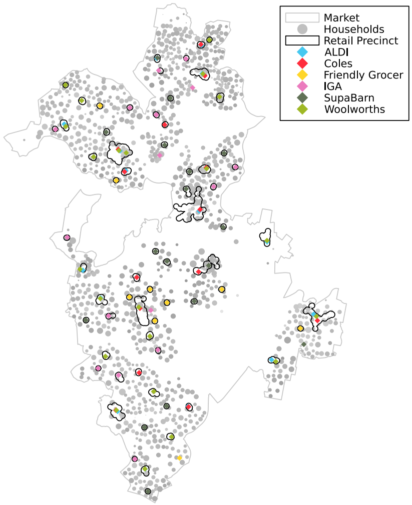


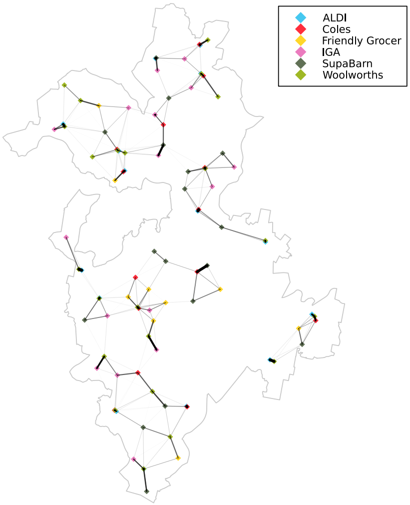
Post: [IGA Woolworths SupaBarn] Karabar



Counterfactuals

Woolworths-IGA Karabar - Geographic Market Definition - Diversion Network



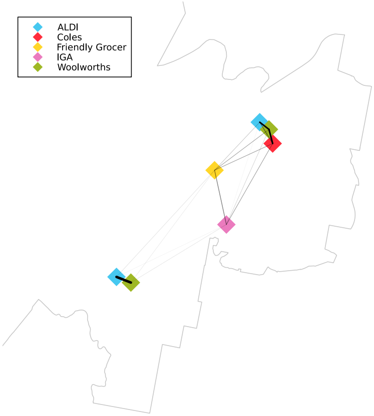


Counterfactuals

Woolworths-IGA Karabar - Pre-Merger - Local Diversion Ratios

[IGA Woolworths SupaBarn] Karabar

$From \backslash To$	AJ	AQ	CQ	FGQ	IK	WJ	WQ
ALDI Jerrabomberra	-1.00	0.00	0.00	0.06	0.02	0.82	0.00
ALDI Queanbeyan	0.00	-1.00	0.26	0.09	0.01	0.00	0.61
Coles Queanbeyan	0.00	0.19	-1.00	0.12	0.08	0.00	0.55
Friendly Grocer Queanbeyan	0.05	0.14	0.25	-1.00	0.18	0.04	0.26
IGA Karabar	0.03	0.03	0.29	0.29	-1.00	0.06	0.14
Woolworths Jerrabomberra	0.82	0.00	0.00	0.04	0.04	-1.00	0.00
Woolworths Queanbeyan	0.00	0.37	0.47	0.10	0.03	0.00	-1.00



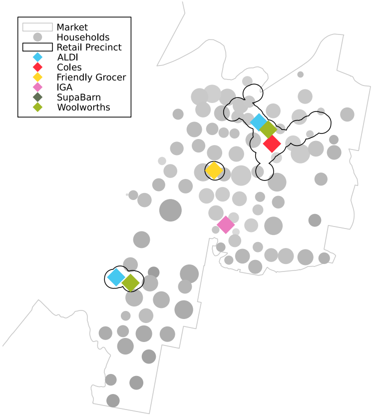
- Diversion ratios: $D_{ij} = (\frac{\partial q_j}{\partial p_i}) / (|\frac{\partial q_i}{\partial p_i}|)$
  - Proportion of customers who leave *i* that switch to *j*
- Localized competition
  - $\widehat{\alpha} a_r + \widehat{\beta}_b \in [10, 15]$
  - $\widehat{\delta} = -4.12$
  - Extreme taste shock/price differential required to travel 3km+
  - Queanbeyan to Jerrabomberra: 6km

Counterfactuals

Woolworths-IGA Karabar - Status Quo - Local Prices & Household Welfare

[IGA Woolworths SupaBarn] Karabar

s	p_s	c_s	q_s	π_s
ALDI Jerrabomberra	2.69	1.00	2192	595
ALDI Queanbeyan	2.24	1.00	2741	280
Coles Queanbeyan	2.48	1.02	4590	1257
Friendly Grocer Queanbeyan	3.96	1.57	3460	2382
IGA Karabar	5.64	2.98	2348	3155
Woolworths Jerrabomberra	4.57	1.29	4245	9155
Woolworths Queanbeyan	2.71	1.29	5231	2681



- Household welfare: $\widehat{W}_h$ =156,634
  - $\widehat{W}_h = \sum_h n_h |\eta_h|^{-1} \ln[1 + \sum_b \Sigma_{bh}^{1-\rho}]$
- IGA Karabar: highest prices
  - Highest marginal costs
  - Local market power: $\widehat{\delta} = -4.12$

Counterfactuals

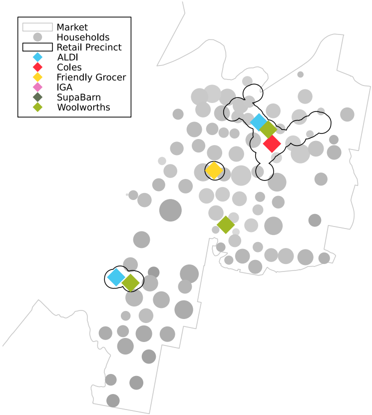
Woolworths-IGA Karabar - Woolworths Acquisition - Local Prices & Household Welfare

[IGA Woolworths SupaBarn] Karabar

s	p_s	c_s	q_s	π_s
ALDI Jerrabomberra	2.72	1.00	2261	786
ALDI Queanbeyan	2.25	1.00	2794	387
Coles Queanbeyan	2.51	1.02	4586	1365
Friendly Grocer Queanbeyan	4.01	1.57	3390	2390
Woolworths Karabar	4.65	1.29	2739	4437
Woolworths Jerrabomberra	4.65	1.29	4164	9246
Woolworths Queanbeyan	2.76	1.29	4892	2439

- Household welfare: $\widehat{W}_h$ =156,146
  - $\Delta\widehat{W}_h = -488$
- IGA $\longrightarrow$ Woolworths Karabar:
  - Quality decrease: 12.51 $\longrightarrow$ 11.99
  - Marginal cost decrease: 2.98 $\longrightarrow$ 1.29
  - (Store-level) price decrease: 5.64 $\longrightarrow$ 4.65

  - ALDI, Coles, Friendly Grocer price increases
  - Loss of differentiated offering in local area

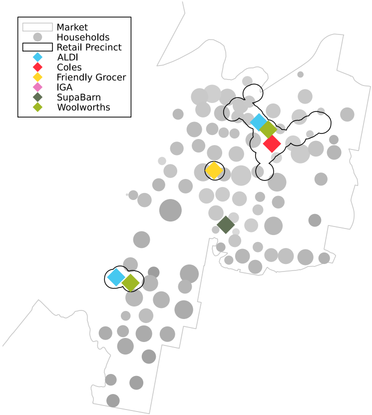


Counterfactuals

Woolworths-IGA Karabar - SupaBarn Acquisition - Local Prices & Household Welfare

[IGA Woolworths SupaBarn] Karabar

s	p_s	c_s	q_s	π_s
ALDI Jerrabomberra	2.68	1.00	2109	419
ALDI Queanbeyan	2.25	1.00	2666	206
Coles Queanbeyan	2.47	1.02	4225	661
Friendly Grocer Queanbeyan	3.86	1.57	3169	1371
SupaBarn Karabar	4.69	1.53	3712	3345
Woolworths Jerrabomberra	4.52	1.29	4169	8722
Woolworths Queanbeyan	2.72	1.29	4971	2345



- Household welfare: $\widehat{W}_h$ =161,429
  - $\Delta\widehat{W}_h = +4,795$
- IGA $\longrightarrow$ SupaBarn Karabar:
  - Quality increase: 12.51 $\longrightarrow$ 12.93
  - Marginal cost reduction: 2.98 $\longrightarrow$ 1.53
  - (Store) price decrease: 5.64 $\longrightarrow$ 4.69

  - No concentration-based price increases
  - No (net) loss of differentiated offering in local area

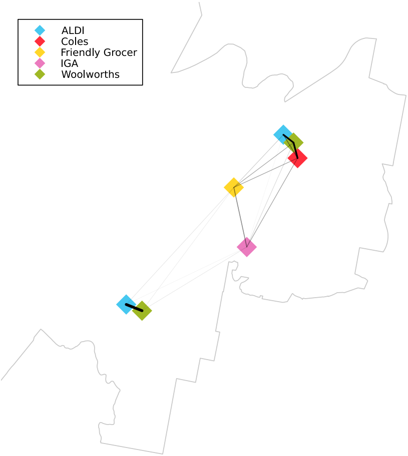
Counterfactuals

Woolworths-IGA Karabar - Pre-Merger - Local Diversion Ratios - δ Sensitivities

$$\delta = [\text{-4.12} \quad \text{-3.00} \quad \text{-2.00} \quad \text{-1.00}]$$

From\To	*AJ*	*AQ*	*CQ*	*FGQ*	*IK*	*WJ*	*WQ*
ALDI Jerrabomberra	-1.00	0.00	0.00	0.06	0.02	0.82	0.00
ALDI Queanbeyan	0.00	-1.00	0.26	0.09	0.01	0.00	0.61
Coles Queanbeyan	0.00	0.19	-1.00	0.12	0.08	0.00	0.55
Friendly Grocer Queanbeyan	0.05	0.14	0.25	-1.00	0.18	0.04	0.26
IGA Karabar	0.03	0.03	0.29	0.29	-1.00	0.06	0.14
Woolworths Jerrabomberra	0.82	0.00	0.00	0.04	0.04	-1.00	0.00
Woolworths Queanbeyan	0.00	0.37	0.47	0.10	0.03	0.00	-1.00

- ACCC analysis consistent with smaller disutility of distance, δ
  - Increased IGA Karabar diversion to/from Queanbeyan/Jerrabomberra stores
  - Denser diversion network
  - Concentration from Woolworths acquisition more harmful



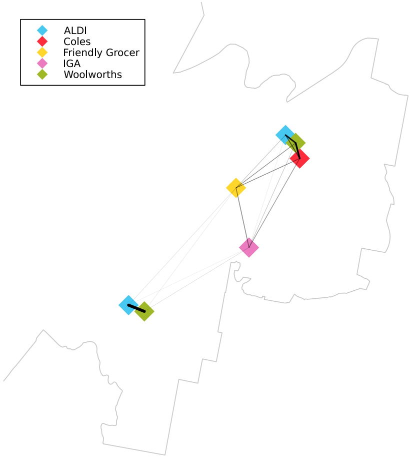
Counterfactuals

Woolworths-IGA Karabar - Pre-Merger - Local Diversion Ratios - δ Sensitivities

$$\delta = \begin{bmatrix} -4.12 & -3.00 & -2.00 & -1.00 \end{bmatrix}$$

From\To	*AJ*	*AQ*	*CQ*	*FGQ*	*IK*	*WJ*	*WQ*
ALDI Jerrabomberra	-1.00	0.00	0.01	0.07	0.03	0.86	0.00
ALDI Queanbeyan	0.00	-1.00	0.32	0.12	0.02	0.00	0.54
Coles Queanbeyan	0.00	0.22	-1.00	0.15	0.09	0.00	0.53
Friendly Grocer Queanbeyan	0.05	0.15	0.29	-1.00	0.16	0.04	0.29
IGA Karabar	0.04	0.06	0.32	0.30	-1.00	0.07	0.20
Woolworths Jerrabomberra	0.86	0.00	0.01	0.05	0.06	-1.00	0.00
Woolworths Queanbeyan	0.00	0.33	0.48	0.13	0.05	0.00	-1.00

- ACCC analysis consistent with smaller disutility of distance, δ
  - Increased IGA Karabar diversion to/from Queanbeyan/Jerrabomberra stores
  - Denser diversion network
  - Concentration from Woolworths acquisition more harmful



Counterfactuals

Woolworths-IGA Karabar - Pre-Merger - Local Diversion Ratios - δ Sensitivities

$$\delta = \begin{bmatrix} -4.12 & -3.00 & -2.00 & -1.00 \end{bmatrix}$$

$From \backslash To$	AJ	AQ	CQ	FGQ	IK	WJ	WQ
ALDI Jerrabomberra	-1.00	0.00	0.03	0.07	0.04	0.84	0.01
ALDI Queanbeyan	0.00	-1.00	0.36	0.13	0.03	0.00	0.48
Coles Queanbeyan	0.01	0.24	-1.00	0.16	0.07	0.01	0.51
Friendly Grocer Queanbeyan	0.04	0.17	0.32	-1.00	0.11	0.04	0.32
IGA Karabar	0.06	0.08	0.29	0.24	-1.00	0.10	0.23
Woolworths Jerrabomberra	0.82	0.00	0.03	0.07	0.07	-1.00	0.01
Woolworths Queanbeyan	0.00	0.30	0.48	0.16	0.05	0.00	-1.00

- ACCC analysis consistent with smaller disutility of distance, δ
  - Increased IGA Karabar diversion to/from Queanbeyan/Jerrabomberra stores
  - Denser diversion network
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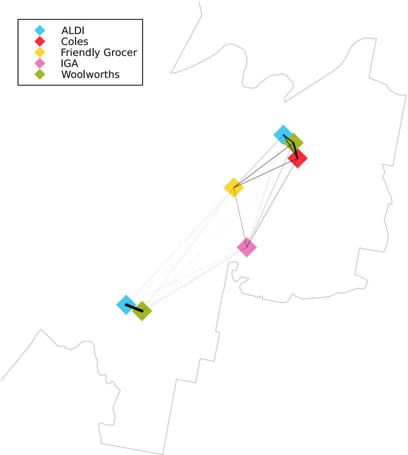

Counterfactuals

Woolworths-IGA Karabar - Pre-Merger - Local Diversion Ratios - δ Sensitivities

$$\delta = \begin{bmatrix} -4.12 & -3.00 & -2.00 & -1.00 \end{bmatrix}$$

From\To	AJ	AQ	CQ	FGQ	IK	WJ	WQ
ALDI Jerrabomberra	-1.00	0.02	0.12	0.10	0.05	0.66	0.04
ALDI Queanbeyan	0.01	-1.00	0.40	0.12	0.02	0.01	0.45
Coles Queanbeyan	0.03	0.26	-1.00	0.15	0.04	0.04	0.49
Friendly Grocer Queanbeyan	0.05	0.17	0.32	-1.00	0.04	0.08	0.33
IGA Karabar	0.09	0.10	0.26	0.14	-1.00	0.16	0.25
Woolworths Jerrabomberra	0.57	0.02	0.15	0.13	0.08	-1.00	0.05
Woolworths Queanbeyan	0.01	0.29	0.50	0.15	0.03	0.01	-1.00

- ACCC analysis consistent with smaller disutility of distance, δ
  - Increased IGA Karabar diversion to/from Queanbeyan/Jerrabomberra stores
  - Denser diversion network
  - Concentration from Woolworths acquisition more harmful